

Airbnb Price Prediction Report - Lyon

Generated on 2026-08-25 by the automated pipeline (*make all*).

Executive summary

The pipeline trained and compared Random Forest, XGBoost and TabPFN on Airbnb listing data for **Lyon** (3798 listings). The **xgboost** model was selected for final evaluation on a held-out test set and achieved: - **RMSE: \$49** - **MAE: \$30** - **R²: 0.70**

Data

- Source: InsideAirbnb (latest available snapshot). See: <https://insideairbnb.com/get-the-data/>
- City: Lyon · 3798 listings after cleaning

EDA

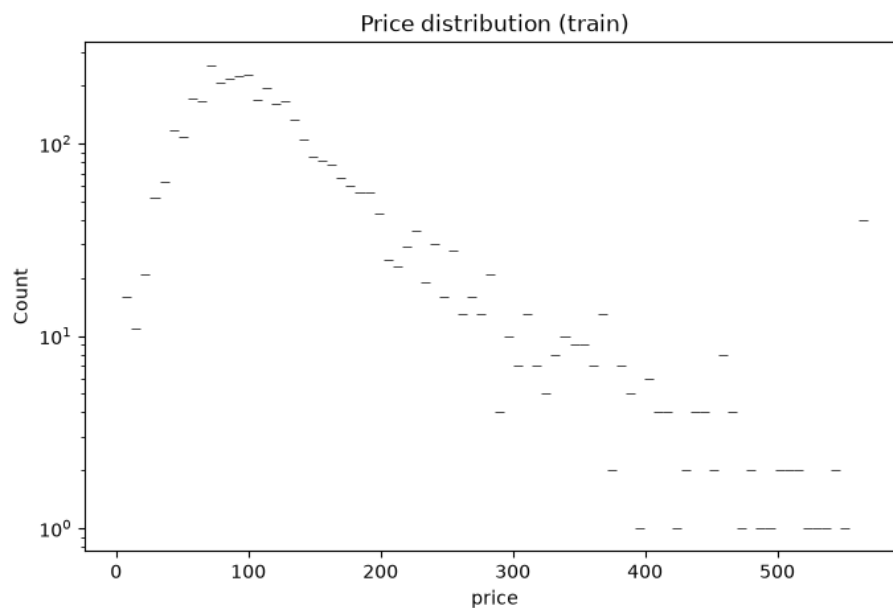
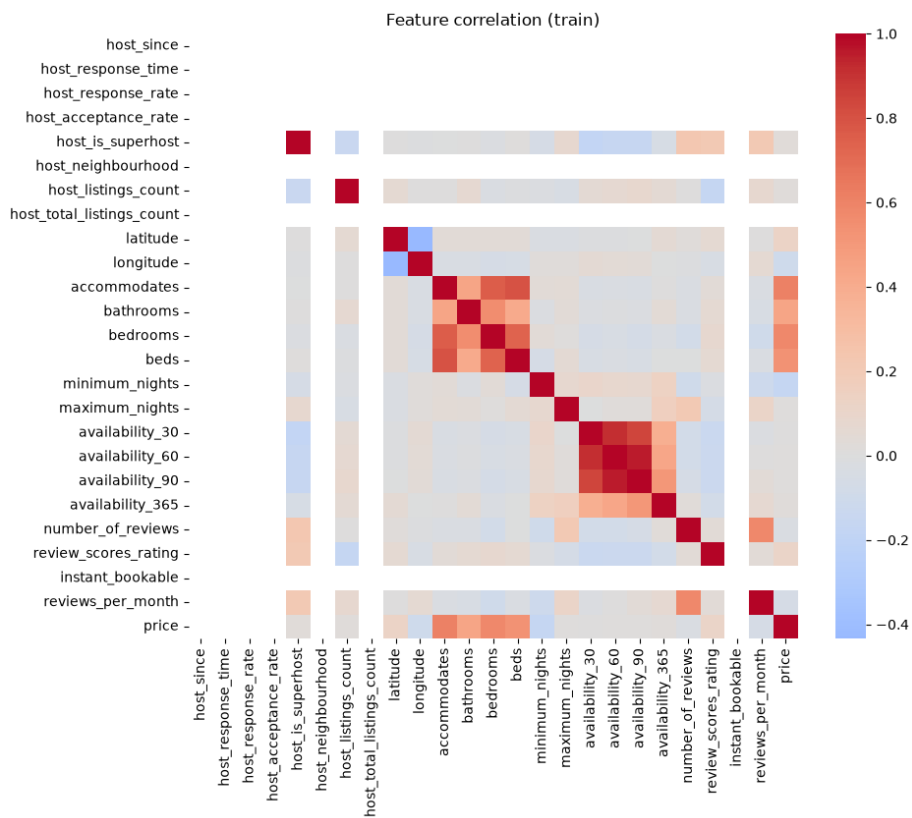
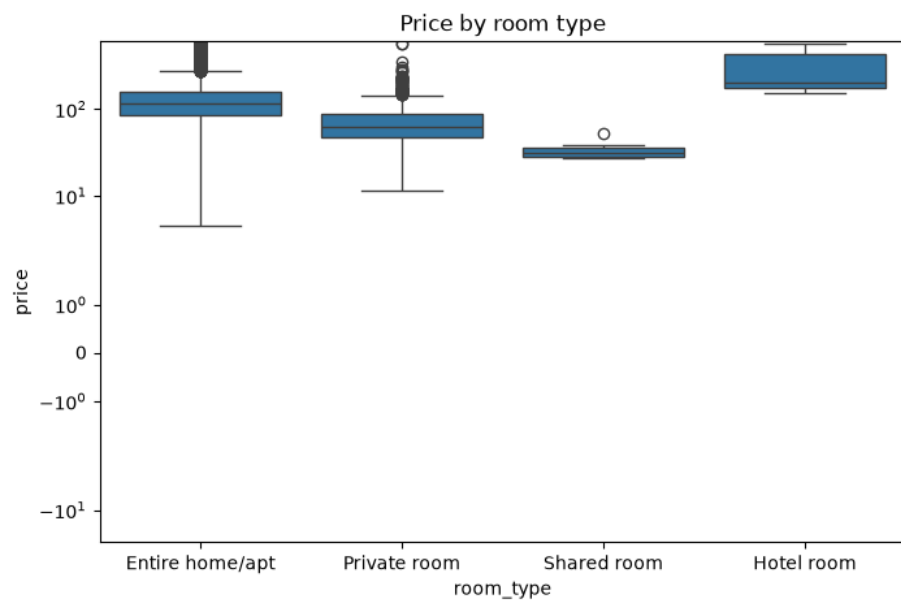


Figure 1: price_dist

- Median price: **\$105** (mean \$129)



Models & Results

Model	Validation RMSE
Random Forest	\$57
XGBoost	\$52
TabPFN (capped sample, not directly comparable)	\$49

Holdout results (best model): see `artifacts/lyon/holdout_results.csv`.

Top features

Feature	Importance
bedrooms	0.180
bathrooms	0.090
accommodates	0.086
room_type_Private room	0.071
minimum_nights	0.059
property_type_Private room in loft	0.057
property_type_Private room in rental unit	0.057
room_type_Hotel room	0.049
property_type_Room in hotel	0.031
room_type_Entire home/apt	0.026

Recommendations

- The best-performing model was **xgboost** (holdout RMSE \$49, MAE \$30, R^2 0.70).
- On validation, **XGBoost** beat Random Forest by 8% on RMSE (\$52 vs \$57). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **bedrooms, bathrooms, accommodates** — i.e. bedrooms (the number of bedrooms); bathrooms (a key listing attribute); accommodates (how many guests a listing sleeps). Hosts can use these levers when setting nightly rates.
- Accuracy caveat: the holdout RMSE of \$49 is **46% of the median listing price (\$105)**, so predictions for typical listings are expected to be off by roughly $\pm\$49$.
- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

Automated pipeline: fetch → preprocess → EDA → train → evaluate → report.
*Code in **src/**, config in **configs/base.yaml**.*